

View results

Respondent

61

Anonymous

68:26

Time to complete

A. General information about the respondent

1. * Country

Luxembourg



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Research and Higher Education (MESR)

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://mesr.gouvernement.lu/fr.html>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The ministry of Research and Higher Education defines, coordinates and implements the national public research policy. It is also responsible for organising, coordinating and overseeing the Luxembourg higher education landscape. It is the supervising ministry of the University of Luxembourg, the public research institutes Luxembourg Institute of Science and Technology (LIST), Luxembourg Institute of Health (LIH) and Luxembourg Institute of Socio-Economic Research (LISER) and of the National Research Fund (FNR).

7. * Full name of officially designated contact person(s) who completed this survey

Stéphanie Schott

8. * Position of officially designated contact person(s) who completed this survey

Head of Department of International Relations

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Romain Martin

11. Brief description of the consultation process established for the preparation of the report

All the public research institutions in Luxembourg as well as the Luxembourg National Data Service and the National Library of Luxembourg were contacted in the framework of a written consultation to provide input for the report.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Luxembourg National Research Fund (FNR), University of Luxembourg, Luxembourg Institute of Science and Technology (LIST), Luxembourg Institute of Health (LIH), Luxembourg Institute of Socio Economic Research (LISER), Luxembourg National Data Service (LNDS), National Library of Luxembourg

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey (Please use commas to separate multiple links, as: link A, link B)

<https://www.fnr.lu/>; <https://www.uni.lu/en/>; <https://www.list.lu/>; <https://www.lih.lu/en/>;
<https://www.liser.lu/>; <https://www.lnds.lu/>; <https://bnl.public.lu/en.html>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☒

Public

☐

Private

☐

Civil Society

☐

Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒

Yes

☐

No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Research and Higher Education, Luxembourg National Research Fund (FNR)

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

French, German

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (*ref.: (i) 16.f, 16.g, 16.h*)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

A number of awareness raising activities were organised by the National Research Fund (FNR), including several workshops, meetings with stakeholders and PhD training sessions. Open science is also a major topic in the FNR's annual funding programmes webinars. The University of Luxembourg, as only public university in the country, and other public research institutions in Luxembourg (Luxembourg Institute of Science and Technology - LIST, Luxembourg Institute of Socio-Economic Research - LISER, Luxembourg Institute of Health - LIH) are involved in various activities that relate to Open Science, in collaboration and individually. □ Nationally through the Luxembourg National Data Service (LNDS) focus group with a concentration on determining synergies and strategy for data in Luxembourg and their Community of Practice, with focus on sharing best practice. □ Nationally through the Consortium of Luxembourg, under the leadership of the National Library of Luxembourg, where the libraries of the four research institutions partner on various digital projects, with some aspects relating to open science through for example transformative agreements etc. □ Nationally through various partnerships and collaborations, e.g. Publications Office of the European Commission, partner in events related to open science aspects, like the Open Data Days in March 2025. □ At the University of Luxembourg, there are institutional efforts e.g: o The Open Science working group, with a mandate from the rectorate, aiming to advance research data management and open science principles and practices., e.g. through trainings, awareness raising and support. o Training, induction activities and awareness raising, undertaken by the above group and dedicated staff at the Luxembourg Learning Centre, namely PhD inductions, PhD workshops, knowledge transfer and intellectual property training for PhD students, induction to new researchers, awareness events during International open access week (Oct) and Love My Data week (March) o The UL Research Data Management Community of Practice ('lunchclub') open to all with an interest in research data

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Support of Open Science, as well as the anchoring of science in society with the goal of establishing Luxembourg as a leading knowledge-based society, is enshrined in the National Research and Innovation Strategy, adopted by Government in 2019. To advance towards the aim of making Luxembourg an economy based on data-driven science and innovation strategies hinges, access to data is crucial. Luxembourg has made significant progress on this front. The Luxembourgish Government's Open Data Strategy (2022) emphasises data re-use and points to the National Data Portal, data.public.lu launched in 2016, as (1) the central syndicate of various national thematic data portals, including those on geospatial data, statistics, mobility, meteorological and environmental data, and (2) as the connection to the European Open Data Portal. In the multi-annual performance agreements (currently for the period 2022-2025) signed between the State and the public research institutions in Luxembourg, the latter commit to implementing a national Open Access policy. The University of Luxembourg's library, the Luxembourg Learning Centre, has had a dedicated post for scholarly communication and open science since 2012, when the institutional repository ORBilu was launched. The role aims to support the needs of researchers in scholarly communication throughout the research cycle through information, training and policy making.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

National Research and Innovation Strategy:
<https://mesr.gouvernement.lu/en/publications/2020/national-research-and-innovation-strategy.html>, Ministry of Higher Education and Research, adoption by Government in 12/2019, publication early 2020

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☐ open science infrastructure
- ☒ open engagement of societal actors
- ☒ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☒ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

A National Open Access Policy was adopted in 2015. The National Research Fund FNR has an Open Access Policy, which is fully aligned with the 2021 Recommendation on Open Science and strongly encourages Open Access Publishing. The Policy was revised in January 2025 due to concerns regarding the high costs related to Open Access publishing. While Open Access publishing was previously mandatory for publications resulting from FNR funded projects, FNR has aligned its policy to the ones of research-performing institutions and does not impose an Open Access obligation where this is not a requirement by the institution the concerned researcher is affiliated to. Relevant documents can be found under the following link: <https://www.fnr.lu/open-science/>

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☒ Yes
- ☐ Under development
- ☐ No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

In the framework of its aforementioned Open Access Policy, the FNR has set up an Open Access Fund to promote the free access to research results from FNR-(co)funded projects by providing financial support to help cover article processing charges. FNR also requires a Data Management Plan (DMP) to be drafted for all projects accepted for funding and has set up related guidelines and templates. In the framework of its missions, the FNR also implements Science in Society activities and events aiming at anchoring science in society and promoting science in school and among the public.

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (*ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h*)

- ☒ Yes
- ☐ Under development
- ☐ No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

Please see elements provided under question 2.2

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (*ref.: (ii) 17.d, 17.e, 17.g*)

- ☒ Yes
- ☐ Under development
- ☐ No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

FNR conducts numerous Science in Society activities with the aim of anchoring science in society and fostering societal impact. The University of Luxembourg has introduced the ORBilu deposit mandate in 2012, which enforces the University's commitment to the open access agenda when it comes to research output. It is mandatory for researchers to deposit scientific peer reviewed articles and published conference proceedings as open access. It also encourages researchers to deposit their selected research output of any type, e.g. if possible open access or at least with a comprehensive metadata record of the reference. The University of Luxembourg has also joined the Coalition for Advancing Research Assessment. Its CoARA action plan includes streamlining of open science practice, supporting the use of ORBilu for ensuring that research output is archived, preserved and visible by the 'as open as possible, as closed as necessary' approach and supporting 'science for society' initiatives through outreach and other citizen science efforts, via the mandated Citizen Science group and DESCOM in particular.

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☒ Yes
- ☐ Under development
- ☐ No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

As previously mentioned, FNR has set up the FNR Open Access Fund which provides financial support to help cover article processing charges for FNR-funded projects. As part of consortium licences to which public research institutions and the National Library of Luxembourg belong, transformative agreements with different publishers have been concluded. Moreover, the National Library of Luxembourg has launched a study in January 2025 to analyse the feasibility of a new national scientific collection at the National Library, bringing together all scientific publications. This study is funded by the FNR. The FNR PSP (Promoting Science to the Public) funding programme aims to bring science closer to society and funds projects that allow the target audience to come into contact with science and research interactively. The FNR RESCOM funding programme aims to support communication in view of disseminating and exchanging scientific outcomes.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

See answer under 2.3.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

40. 2.8 cont. **If yes**, please provide details and links as relevant.

In the framework of FNR funded public-private partnerships, projects must align with the regular recommendations.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☐ Yes

☒ Under development

☐ No

42. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

There is no monitoring framework, however in the multi-annual performance agreements signed between the State and the public research institutions in Luxembourg, the latter commit to implementing a national Open Access policy. The University of Luxembourg's open science working group takes part in UNESCO working groups for open science, e.g., developing a monitoring initiative and developing indicators for measurement of open science progress.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

Please refer to data by Eurostat (latest data available for 2023):
https://ec.europa.eu/eurostat/databrowser/view/rd_e_gerdtot_custom_15368188/bookmark/table?lang=en&bookmarkId=33cea9c8-d43e-4cbf-a4a9-bb7e32f444d7

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

In 2022, 93,3% of households had access to Fixed Very High Capacity Network Coverage (source: Digital Decade Country Report Luxembourg 2023, published in September 2023)

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

The Restena Foundation interconnects and provides network and security services for research and education institutions in Luxembourg, as well as in the health, culture and national administration sectors.

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☒ community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Bibliothèque nationale du Luxembourg digital library, accessible for members of the different institutions of the country as well as for private persons, offering a broad variety of publications, books; <https://bnl.public.lu/en/offres-numeriques/a-z.html>; ORBilu: University of Luxembourg Open Repository and Bibliography Melusina Press: the University of Luxembourg diamond press; <https://www.melusinapress.lu/> Research Portal Luxembourg Institute of Health (LIH) : providing access to publications/research data, openly accessible in the internet via the LIH webpage: <https://researchportal.lih.lu/> Research Portal Luxembourg Institute of Socio-Economic Research (LISER) : LISER publishes a wide series of publications available to the public via its PURE research portal, i.e. some 2,700 publications in open access and also available via sites such as RePEc.org (Research Papers in Economics) and SSRN.com (Social Science Research Network): <https://liser.elsevierpure.com/> Luxembourg National Data Service (LNDS): organisation created by the government to support the secondary use of public sector data, in a trustable manner. As part of its services the LNDS facilitates the Data Management National Community of Practice. The community is a platform for exchange of knowledge and best practice on the topics of FAIR data management and sharing. This community is open to the participation of all individuals with a professional interest in data management and stewardship. Luxembourg Open Data platform: <https://data.public.lu/fr/>

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

51. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Luxembourg hosts the Meluxina supercomputer (<https://www.luxprovide.lu/meluxina/>) as part of the EuroHPC Joint Undertaking (JU) initiative. EuroHPC access policies require the practice of Open Science as a condition of access to the computing facilities.

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

54. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

The Open Science working group at the University of Luxembourg collaborates with other researchers and support staff in delivering training at least twice a year to PhD students in open science, open access, copyright, intellectual property, data management etc., through lectures and workshops. The PhDs receive ECTS for attending these training sessions, which run in conjunction with the events listed below. The Luxembourg Learning Centre supports students and researchers with training around information literacy, ethical use of information and open science and practical sessions with librarians learning new skills. UL also holds informal sessions twice a year with lectures, activities, competitions and fun for all students, researchers and staff, i.e.: o International open access week in October o Love my data week in March The Luxembourg National Data Service (LNDS) has recently joined the Skills4EOSC network of competence centres. At the Luxembourg Institute of Science and Technology (LIST), there is a dedicated team on Research Data Management since January 2025. The team embraces the role of research data stewards. As such, they will implement and support a state-of-the art management of LIST's digital research output and foster internal and external collaboration and reuse of data. This includes among others individual researcher and project support, providing Open Science/ Research Data Management (RDM) expertise and consultancy, FAIR Data Management Planning and data curation support, developing and supporting RDM tools, training on best practices in RDM, scientific data and metadata management, long-term and person-independent archiving of research results and representing LIST in national and international consortia. The Luxembourg Institute of Socio-Economic Research (LISER)'s IS division provides accredited and trusted tools and services that enable both data producers and data users to easily share, find and access reusable and interoperable data. This in turn responds to the growing needs of open science, while respecting the security, integrity and personal data privacy of the information at stake. In addition, LISER is driving the development of Open Science at the Institute, and this topic is fully integrated in the annual institutional training plan which offers trainings related to this topic to its researchers and staff at large. Special attention is paid to socio-economic surveys and databases, whose information is extremely interesting for research, but often contains personal data that are handled in accordance with the GDPR. However, as far as possible, data and research results are made available to the general public through agreements between LISER and its partners.

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☒ Yes

☐ No

58. 4.3 cont. **If yes**, please provide more information and links as relevant.

see answer to 4.1 above

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☒ Partly
- ☐ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

61. 4.5 cont. **If yes**, please provide more information and links as relevant.

• FNR Science in Society programme: <https://www.fnr.lu/science-in-society-2/> • FNR Pairing Scheme: Politics Meets Research: <https://www.fnr.lu/pairing-scheme-politics-meets-research/> • SciCom Guide ("50 essentials on science communication"), published by FNR and University of Luxembourg: <https://www.degruyter.com/document/doi/10.1515/9783110763577/html> • Dedicated scholarly communication librarian specialist in the Luxembourg Learning Centre to support researchers and the University of Luxembourg community in open science topics, for example communicating research throughout the research cycle, publishing, sharing data, writing a data management plan, depositing in ORBilu etc. • Doctoral education in Science Communication at the University of Luxembourg: <https://www.uni.lu/life-en/descom/>

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

63. 5.1 cont. **If yes**, please provide more information and links as relevant.

FNR as well as the University of Luxembourg have signed the DORA declaration (San Francisco Declaration on Research Assessment) as well as the CoARA (Coalition for Advancing Research Assessment) agreement. The University of Luxembourg has adopted a CoARA action plan (<https://zenodo.org/records/10560904>). To drive the internal reflection process on these topics, Special Advisors for Research Assessment and for Open Science and Data Management have been appointed at the University of Luxembourg.

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☒ Yes

☐ No

65. 5.2 cont. **If yes**, please provide details and links as relevant.

According to the Law of 27 June 2018 on the organisation of the University of Luxembourg (<https://legilux.public.lu/eli/etat/leg/loi/2018/06/27/a587/jo>), the tasks of professors at the University comprise teaching, research, knowledge transfer and dissemination of research results, national, European and international cooperation as well as administration and management. These different aspects are equally valued and are considered in hiring and promotion procedures. • FNR has implemented the narrative CV in the framework of its funding application procedures, which includes Open Science aspects.

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☒ Yes

☐ No

67. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders^[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

Please see response to question 5.2.

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.



Yes



No

69. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

Open access publishing has led to significant increases in publication costs for research institutions due to article processing charges. As a result, the FNR Open Access Policy was adapted, as mentioned under question 2.2., to allow for more flexibility in publishing choices in respect of research institutions' and researchers' autonomy, while continuing to promote Open Access publishing and Open Science practices.

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

70. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☒ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☐ No

71. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☒ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☐ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☒ The Luxembourg Institute of Socio-Economic Research (LISER) has a dedicated Competence

72. 6.1. cont. **If yes**, please provide more information and links as relevant

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

73. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Membership in European Open Science Cloud Association; FNR membership of cOAlition S, an international consortium of research funding and performing organizations that implement the Open Access initiative Plan S in a coordinated way FNR and the Luxembourg Institute of Socio-Economic Research (LISER) are members of ORCID FNR is a signatory of the Berlin Declaration on Open Access to Knowledge in the Sciences and Humanities FNR and the University of Luxembourg have signed the DORA declaration (San Francisco Declaration on Research Assessment) FNR and the University of Luxembourg have signed the CoARA (Coalition for Advancing Research Assessment) agreement. The University of Luxembourg has adopted a CoARA action plan. National collaboration of libraries through the Consortium of Luxembourg National best practice sharing and determining of synergies and strategy for data through the Luxembourg National Data Service (LNDS) focus group Community of Practice Participation in ELIXIR, the European infrastructure for life science information, through the Luxembourg Elixir node Involvement by University of Luxembourg in UNESCO Open Science monitoring working group (WG3 Open science monitoring with scholarly content providers and WG1 Scoping the Needs of Open Science Monitoring Involvement by University of Luxembourg in EARMA Open Science community of practice University of Luxembourg collaboration with the universities of Mons and Liège (Belgium) on the institutional repository ORBi/ORBilu (archive for research publications) Greater Region collaboration on open science between University of Luxembourg, University of Saarbrücken and University of Lorraine (informal discussions and sharing of best practice)

74. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

75. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific plan or strategy, but active multi-stakeholder collaboration and engagement at national, regional and international level through diverse organisations and memberships, as mentioned above.

76. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

77. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 8. GENERAL CONSIDERATIONS

78. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

79. 8.1 cont. **If yes**, please provide more information.

High costs related to Open Access publishing.

80. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org